PROTOCOL for Management of Patients with Acute Ischemic Stroke

Emergency Evaluation and Diagnosis of Acute Ischemic Stroke

The organized protocol for the emergency evaluation of patients with suspected stroke is as given below .
The goal is to complete an evaluation and to begin intravenous thrombolytic therapy treatment within 60 minutes of patient's arrival in an emergency department making the best of efforts to administer this within one hour of onset of stroke symptoms after haemorrhage is ruled out by CT scan brain, but certainly not later than 4.5 hours (270 minutes) f onset of stroke symptoms. The acute stroke team includes Emergency Medicine physicians, nurses, and laboratory/radiology personnel and stroke on call neurologist. Every stroke patient should have a careful clinical assessment, including neurological examination.

Documentation of detailed but focussed history of onset and progress, past TIAs and strokes, BP, Pulse rate and rhythm, level of sensorium, and a stroke rating scale, preferably the National Institutes of Health Stroke Scale (NIHSS) are mandatory in the initial evaluation and documentation.

A limited number of hematologic (CBC), coagulation (PT/INR) and Biochemistry tests (BUSE) and CT Brain are done during the initial emergency evaluation, but only blood glucose and CT brain are mandatory investigations for the initiation of intravenous rtPA. In intravenous fibrinolysis candidates, the brain imaging study should be done and interpreted within 30 minutes of patient arrival in the emergency department by a physician with expertise in reading CT and MRI studies of the brain

Baseline electrocardiogram is recommended in patients presenting with acute ischemic stroke but should not delay initiation of intravenous rtPA. CT angiogram/Diffusion MRI, Baseline troponin, chest radiographs and ECHO are also recommended but should not delay initiation of intravenous rtPA

A. Upto 3 hrs: During stroke on call neurologist's (acute stroke team member) evaluation, those presenting within 2.5 hrs of onset of confirmed non-haemorrhagic stroke, will be screened for open label, single arm Tenecteplase thrombolysis trial (to be administered within 3 hours) and all eligible patients will be offered the opportunity of participating in this national CTRI registered and JIPMER IEC approved multicentric trial and those giving informed consent will be recruited to IV Tenecteplase trial. They will be admitted into Acute stroke unit (148A)
B. Upto 4.5 hrs: Those ineligible or not consenting for Tenecteplase trial but within 4.5 hrs of onset of stroke will be screened for eligibility for IV tPA and if found eligible administered IV tPA. They will be admitted into Acute stroke unit (148A)
C. Upto 6-12 hrs: Those ineligible for Tenecteplase trial or IV tPA administration but within 6 hrs of anterior circulation ischemic stroke or within 12 hrs of posterior circulation ischemic stroke, will be considered for IA tPA and after consent will be treated with IA tPA. They will be admitted into Acute stroke unit (148A)
D. Upto 24 hrs: Those ineligible or not consented for Tenecteplase trial or IV tPA or IA tPA but within 24 hrs of ischemic stroke, will be screened for EDOZ neuroprotection trial and if eligible will be offered opportunity to participate and those consenting will be recruited. They will be admitted into Acute stroke unit (148A) or Neurology ward (148)
E. Currently acute ischemic stroke mechanical thrombectomy is not being done in JIPMER Those ischemic strokes, not recruited or excluded from EDOZ trial and developing features of significant mass effect, will be subjected to measures to lessen the risk of edema and close monitoring of the patient for signs of neurological worsening during the first days after

stroke. Placement of a ventricular drain is done for patients with acute hydrocephalus secondary to ischemic stroke. They are considered for surgical decompression of anterior or posterior cranial fossa and those eligible will be referred to Neurosurgery for screening for the same and will be subjected to decompression on consent. They will be admitted into Acute stroke unit (148A)

F. Those ischemic stroke patients with cardioembolism from confirmed cardiac thombus on ECHO or have carotid artery dissection will be given IV heparin therapy
G. All ischemic strokes not receiving IV tenecteplase or IV tPA or IV heparin will routinely receive Aspirin 325 mg orally if no evidence for any acute upper GI or other haemorrhage. Aspirin can be given only 24 hours after administration of thrombolytic drugs. Those on earlier antiplatelet medications will increased dosage upto 325, in case of aspirin followed by 150 mg daily; those on clopidogreal or aspirin plus dipyridamole will continue to receive same and investigated for antiplatelet failure for primary or secondary failure
H. Those ischemic strokes arriving EMS, after 24 hours of stroke onset or not under clauses A-G will be referred to Medicine by the EMS for further evaluation and management. Cases not admitted by Medicine will be cared for my EMS for 48 hrs upto a maximum of 72 hours and thereafter discharged from EMS with appropriate advice to attend stroke follow up clinic of Neurology medicine (Thursday) for further management.
I. Stable patients who need long term physical or cognitive rehabilitation will be assessed by stroke clinic for clearly achievable goals by further admission and admitted to stroke recovery ward (148B). Those recovering from acute stroke unit will also be transferred to Neurology general ward or stroke recovery ward depending on how stable they are.
J. Those admitted for specific goals of secondary stroke prevention such as carotid stenting, carotid endarterectomy and other revascularisation procedures will also be admitted in stroke recovery ward.
K. Those old stroke or acute stroke patients with acute worsening due to exacerbation of COPD or acute heart failure or renal failure or uncontrolled DM or HT or acute multisystem involvement will be admitted and managed by general medicine or pulmonary medicine, cardiology, nephrology or endocrinology until they are stabilised and discharged for appropriate follow up.
L. Statins are continued in acute stroke patients who have already been on statins.
M. Those patients with post stroke uncontrolled seizures with or without fresh deficits; dementia with delirium etc. can be referred to Neurology for further management and will be admitted in acute stroke unit subject to availability of beds
N. Patients with intracerebral haemorrhage or subarachnoid haemorrhage not admitted by Neurosurgery, patients with venous strokes and strokes complicating pregnancy etc. not admitted by Medicine can be referred to Neurology stroke on call resident and will be admitted for management in Acute stroke unit (provided there are no multisystem involvement needing intensive general medicine care) subject to availability of beds in acute stroke unit (148A).

Brain and Vascular Imaging

For patients with acute cerebral ischemic symptoms that have not yet resolved:

Emergency imaging of the brain (normally non-contrast CT scan) before initiating any specific therapy to treat acute ischemic stroke. In most instances, non–contrast-enhanced CT will provide the necessary information to make decisions about emergency management to exclude intracerebral hemorrhage (absolute contraindication) and to determine whether CT hypodensity or MRI hyperintensity suggest potentially large brain infarct. If frank

hypodensity involves more than one third of the middle cerebral artery territory, intravenous rtPA treatment should be withheld.

For patients with cerebral ischemic symptoms that have resolved**: TIA**

Patients with transient ischemic neurological symptoms is subjected to neuroimaging evaluation within 24 hours of symptom onset or as soon as possible in patients with delayed presentations. MRI, including diffusion-weighted imaging, is the preferred brain diagnostic imaging modality. If MRI is not available within this period, head CT is performed. Duplex followed if necessary by MRA imaging of the cervical vessels are performed routinely as part of the evaluation of patients with suspected TIAs to exclude the presence of proximal intracranial stenosis and/or occlusion and should be obtained when knowledge of intracranial steno-occlusive disease will alter management. Catheter angiography is done to confirm abnormalities detected with noninvasive testing

Ischemic stroke: General Supportive Care and Treatment of Acute Complications:

1. Cardiac monitoring is recommended to screen for atrial fibrillation and other potentially serious cardiac arrhythmias that would necessitate emergency cardiac interventions. Cardiac monitoring should be performed for at least the first 24 hours
2. Patients who have elevated blood pressure and are otherwise eligible for treatment with intravenous rtPA should have their blood pressure carefully lowered so that their systolic blood pressure is <185 mm Hg and their diastolic blood pressure is <110 mm Hg before thrombolytic therapy is initiated. If medications are given to lower blood pressure, blood pressure is stabilized at the lower level before beginning treatment with intravenous rtPA and maintained below 180/105 mm Hg for at least the first24 hours after intravenous rtPA treatment. If patient is not receiving tPA, lower blood pressure by 15% during the first 24 hours after onset of stroke but AHT medications withheld unless the systolic blood pressure is >220 mm Hg or the diastolic blood pressure is >120 mm Hg. Antihypertensive therapy within 24 hours of stroke is relatively safe. Restarting antihypertensive medications is reasonable after the first 24 hours for patients who have preexisting hypertension and are neurologically stable. Patients who have malignant hypertension or other medical indications for aggressive treatment of blood pressure should be treated accordingly.
3. Airway support and ventilatory assistance are recommended for the treatment of patients with acute stroke who have decreased consciousness or who have bulbar dysfunction that causes compromise of the airway. Supplemental oxygen should be provided to maintain oxygen saturation >94% .
4. Sources of hyperthermia (temperature >38°C) will be identified and treated, and antipyretic medications should be administered to lower temperature in hyperthermic patients with stroke.
5. Hypoglycemia (blood glucose <60 mg/dL) should be treated in patients with acute ischemic stroke.The goal is to achieve normoglycemia. Hyperglycemia during the first 24 hours: Treat hyperglycemia to achieve blood glucose levels in a range of 140 to 180 mg/dL and to closely monitor to prevent hypoglycemia.
6. Hypovolemia should be corrected with intravenous normal saline, and cardiac arrhythmias that might be reducing cardiac output should be corrected. But Volume Expansion, Vasodilators, and Induced Hypertension have no role
7. Patients with suspected pneumonia or urinary tract infections will be treated with appropriate antibiotics.
8. Subcutaneous administration of anticoagulants for treatment of immobilized patients to prevent deep vein thrombosis when paralysed limbs have a power less than grade 2. Aspirin can be given for treatment of patients who cannot receive anticoagulants for prophylaxis of DVT.Intermittent external compression devices is advised for treatment of patients who cannot receive anticoagulants
9. Assessment of swallowing before the patient begins eating, drinking, or receiving oral medications. Patients who cannot take solid food and liquids orally is given nasogastric, nasoduodenal, or percutaneous endoscopic gastrostomy tube feedings to maintain hydration and nutrition while undergoing efforts to restore swallowing. If swallowing is impaired, nasogastric feeding is preferred to PEG for first 2-3 weeks after stroke.
10. Early mobilization of less severely affected patients and measures to prevent subacute complications of stroke
11. Treatment of concomitant medical diseases
12. Discouraged: Routine use of indwelling catheter, prophylactic antibiotics, nutritional supplments

Intravenous thrombolysis

1. Intravenous rtPA (0.9 mg/kg, maximum dose 90 mg) is recommended for selected patients who may be treated within 4.5 hours of onset of ischemic stroke.
2. In patients eligible for intravenous rtPA, benefit of therapy is time dependent, and treatment should be initiated as quickly as possible. The door-to-needle time (time of bolus administration) should be within 60 minutes from hospital arrival.
3. In the time period of 3 to 4.5 hours after stroke onset, the eligibility criteria for treatment are similar to those for people treated at earlier time periods within 3 hours, with the additional exclusion criteria: patients >80 years old, those taking oral anticoagulants regardless of INR, NIHSS score >25, imaging evidence of ischemic injury involving more than one third of the middle cerebral artery territory, or those with a history of both stroke and diabetes mellitus
4. Those undergoing IV tPA monitored carefully for intracranial and extracranial bleeds and also for angioedema.
5. Use of intravenous fibrinolysis in patients with conditions of mild stroke deficits, rapidly improving stroke symptoms, major surgery in the preceding 3 months, and recent myocardial infarction may be considered, and potential increased risk should be weighed against the anticipated benefits.
6. Intravenous rtPA is not given to patients taking direct thrombin inhibitors, direct factor Xa inhibitors;unless the patient has not received a dose of these agents for >2 days. Can be given to patients on VKA with INR <1.7 , if age is less than 80.). Similar consideration should be given to patients being considered for intra-arterial rtPA.

Endovascular Interventions

1. Patients eligible for intravenous rtPA should receive intravenous rtPA even if intra-arterial treatments are being considered.
2. Rescue intra-arterial fibrinolysis or mechanical thrombectomy for recanalization in patients with large-artery occlusion who have not responded to intravenous fibrinolysis but mechanical thrombolysis using solitaire FR or trevo stent retrievers yet to be established in JIPMER
3. The usefulness of mechanical thrombectomy devices other than the Merci retriever, the Penumbra System, Solitaire FR, and Trevo is not yet established in JIPMER.
4. Intra-arterial thrombolysis also considered in patients with contraindications for IV thrombolysis. IA rtPA dose is normally upto 10 mg but 20 mg is the highest dose.
5. Emergent angioplasty and/or stenting of the extracranial carotid or vertebral arteries not considered in unselected patients but considered in the treatment of acute ischemic stroke resulting from cervical atherosclerosis or dissection